

Wei-Tao Lyu

Curriculum Vitae · September 2026

Postdoctoral Fellow in Physics · Chinese University of Hong Kong

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Hong Kong

RESEARCH INTERESTS

- Microwave Kinetic Inductance Detectors (MKIDs)
- Submillimetre astronomy
- Superconducting resonators
- Cryogenic instrumentation
- Detector readout and calibration

TECHNICAL SKILLS

- **Instrumentation:** Superconducting resonators, MKIDs, device characterization
- **Cryogenics:** Cryostat integration and optimization for astronomy systems
- **Data analysis:** Signal processing and calibration for superconducting devices
- **Software:** Simulation and data analysis tools for instrumentation work

PROFILE

Astronomical instrumentation researcher working on superconducting Microwave Kinetic Inductance Detectors, submillimetre camera systems, cryogenic integration, and readout characterization.

APPOINTMENTS

2022 – Present

Postdoctoral Fellow

Department of Physics, Chinese University of Hong Kong.
Supervisor: Prof. Hua-Bai Li.

2022

Research Associate

Department of Physics, Chinese University of Hong Kong.
Supervisor: Prof. Hua-Bai Li.

EDUCATION

2015 – 2021

PhD in Astronomy

School of Astronomy and Space Science, University of Science and Technology of China. Advisor: Prof. Sheng-Cai Shi.

2011 – 2015

Bachelor of Science in Physics

School of Physics, Huazhong University of Science and Technology.

SELECTED PUBLICATIONS

Laser-Based Frequency-to-Pixel Mapping Method for Submillimeter MKID Array

Lyu, W., Li, H. B., Zhi, Q., Li, J., & Shi, S. (2026). *RAS Techniques and Instruments*, rzag007.

新一代亚毫米波偏振仪Roger

吕伟涛, 黄俊锟, 孙笳淋, 李华白 (2025). *天文学进展*, 第43卷 第4期.

ROGer: Remote Observing from Greenland

Weitao Lyu, Junkun Huang & Hua-bai Li (2024). *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XII*, 131021D.

Disorder effects in NbTiN superconducting resonators

Lyu, W. T., Zhi, Q., Hu, J., Li, J., & Shi, S. C. (2024). *Chinese Physics B*, 33(2), 027401.

Cavity and ground effects on a superconducting microstrip resonator

Lv, W. T., Ding, J. Q., Zhi, Q., Wang, Z., Li, J., & Shi, S. C. (2021). *AIP Advances*, 11(9), 095308.

RESEARCH PROJECT

Hong Kong's Window on the Universe

Building a pioneering submillimetre astronomical camera. Role: cryostat system integration with superconducting MKIDs and cold optics.